ROUND 1 REVIEW

AI Frontier Challenge 2026 Round 1 Submission

Team Name

[Enter Team Name - Placeholder]

Team Leader

[Enter Leader Name - Placeholder]

Track

[Track: Problem H5 - ASHA Companion]

Date

[Date: 18/10/2026 - Placeholder]

Use Case

Problem Statement

- **Frontline Burden & Zero Connectivity:**
  Over 10.29 lakh rural ASHA/ANM workers juggle bulky paper flipbooks and 20+ disconnected apps across 1,700+ remote AP villages with zero mobile reception.
- **Preventable Maternal & Newborn Mortality:**
  Obstetric haemorrhage causes 47% of maternal deaths; missed newborn danger signs (hypothermia, convulsions, poor suckling) lead to tragic delays in referral.
- **LLM Hallucination & Scope Risks:**
  Generic cloud AI models hallucinate plausible but dangerous clinical dosages, cannot enforce non-prescribing scope boundaries, and fail completely offline on low-end phones.

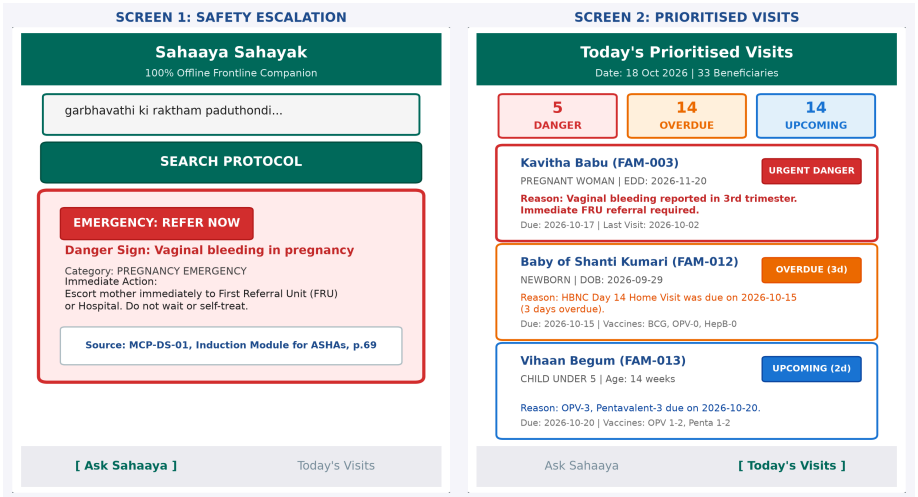
Market Opportunity

- **Target User Scale & Reach:**
  Over 10.29 lakh active ASHAs across India (norm of ~1 per 1,000 population) plus sub-centre ANM nurse-midwives delivering frontline primary care.
- **Fragmented Competitive Landscape:**
  Existing tools (ANMOL, mSakhi, Kilkari, SMARThealth GPT) require cloud connectivity, increase data-entry fatigue, or lack hard-coded emergency escalation.
- **Sahaaya High-Impact Value:**
  A 100% offline, protocol-grounded companion: deterministic danger-sign escalation first, zero cloud dependency, doctor redirects for out-of-scope queries, and prioritised visits.

Proposed Solution

Solution Overview

- **Deterministic Danger Escalation First:**
  Scans queries across English, Hinglish & Telugu (Latin); triggers red 'REFER NOW' screen before any search logic (100% recall on 367 phrasings).
- **Strict Non-Prescribing Scope Gate:**
  Automatically blocks dosage and antibiotic queries with doctor redirects, keeping paramedical workers safely within ASHA guidelines.
- **Extractive BM25 Protocol Retrieval:**
  Pure Kotlin BM25 matches questions to official MoHFW/NHM guidelines with exact document & page citations (0% hallucination).
- **Prioritised Daily Household Action Plan:**
  Parses village registers into an actionable daily route: Danger Flags > Overdue Follow-ups > Nearest Due Dates.



Design Steps & Architecture

Technical Implementation

1. **Hard-Coded Danger Rule Engine:**
   Matches multilingual tokens (EN, Hinglish, Telugu-Latin) for 27 maternal & newborn danger signs. Triggers immediate red REFER NOW card with protocol citations.
2. **Parametrical Intent Safety Gate:**
   Intercepts out-of-scope requests (antibiotics, clinical dosages, medical diagnoses) and returns fixed physician redirect to prevent malpractice.
3. **Pure Kotlin BM25 Retriever:**
   Tokenises query with code-mixed synonym expansion, scores against grounded MoHFW chunks with strict threshold gating to prevent hallucination.
4. **Dynamic Visit List Engine:**
   Ranks 30+ family records by clinical urgency: Danger Flags (Urgent) -> Overdue visits (by days overdue) -> Upcoming routine follow-ups.

Architecture Diagram

Challenges & Mitigation

Challenge 1

Problem: Zero mobile reception in rural hamlets and severe RAM/battery constraints on low-end ASHA smartphones.

Mitigation: 100% on-device architecture with ZERO network permissions. Measured APK size is 15.65 MB, median response latency is 0.53 ms, and peak RAM footprint is < 1 MB.

Challenge 2

Problem: Medical hallucination risks and catastrophic delays from missed pregnancy and newborn danger signs.

Mitigation: Deterministic danger rules execute BEFORE any retrieval logic (achieving 100% recall on 367 verified multilingual phrasings). Answers strictly cite section and page.

Challenge 3

Problem: Code-mixed vernacular queries (English, Hinglish, Telugu in Latin script) with spelling variations and typos.

Mitigation: Engineered robust phonetic tokenisation, code-mixed synonym mapping, and token-proximity matching. On-device SLM rephraser planned for finale using organizer models.

Unique Selling Proposition (USP)

Key Differentiator 1

Deterministic Safety-First Layer:
Hard-coded emergency rule engine fires BEFORE any AI logic, achieving 100% recall (367/367 tested phrasings) with zero hallucination and cited emergency referral instructions.

Key Differentiator 2

100% Offline & Measured Speed:
Zero network permissions required.
Measured on JVM & Android device:
· Median Latency: 0.53 ms
· P95 Latency: 5.38 ms
· APK Size: 15.65 MB | RAM: < 1 MB
Works seamlessly in airplane mode.

Key Differentiator 3

Prioritised Daily Action Plan:
Transforms static paper registers into an automated daily workflow:
1. URGENT: Danger Flag Cases
2. OVERDUE: Ranked by days overdue
3. UPCOMING: Nearest scheduled care
Reduces frontline planning fatigue.

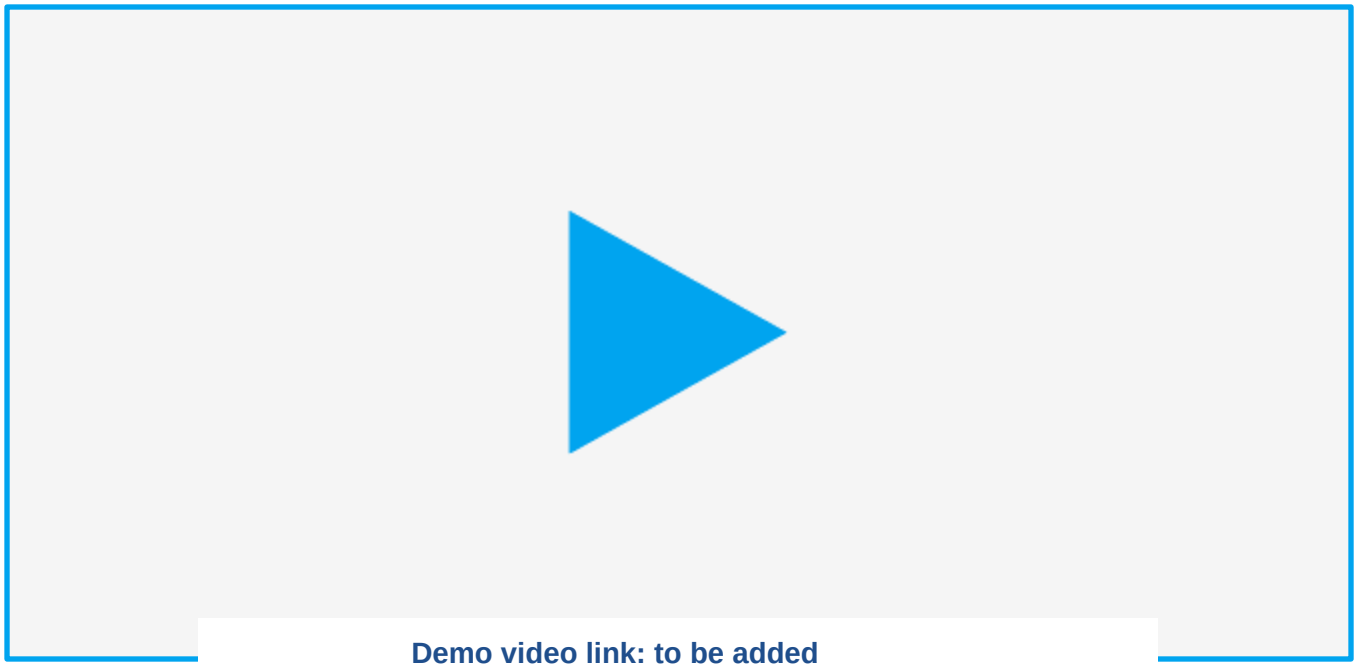
Competitive Advantage Summary

· **Clinically Grounded Trust:**
Unlike generic LLM chatbots that invent antibiotic dosages, Sahaaya strictly refuses prescriptions and cites official MoHFW protocols with exact document and page numbers.

· **Workflow Unification:**
Replaces fragmented flipbooks, complex danger checklists, and multi-app fatigue with a clean two-screen interface tailored for village health workers in low-connectivity areas.

Demo & POC Video

Live Demonstration



Demo video link: to be added

Working Offline Android POC (app-debug.apk, 15.65 MB) | 100% Danger Recall (367 phrasings) | 0.53 ms Median Latency | Zero Cloud Permissions